FIITJEE

JEE(Main)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – VIII
(Main)

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**C**	31.	**C**	61.	**C**
2.	**B**	32.	**B**	62.	**C**
3.	**D**	33.	**B**	63.	**D**
4.	**C**	34.	**C**	64.	**B**
5.	**D**	35.	**B**	65.	**D**
6.	**D**	36.	**C**	66.	**A**
7.	**B**	37.	**B**	67.	**C**
8.	**C**	38.	**B**	68.	**C**
9.	**D**	39.	**B**	69.	**C**
10.	**A**	40.	**A**	70.	**C**
11.	**C**	41.	**A**	71.	**C**
12.	**D**	42.	**A**	72.	**C**
13.	**B**	43.	**B**	73.	**B**
14.	**A**	44.	**A**	74.	**B**
15.	**D**	45.	**B**	75.	**A**
16.	**A**	46.	**A**	76.	**A**
17.	**C**	47.	**B**	77.	**C**
18.	**D**	48.	**B**	78.	**B**
19.	**B**	49.	**A**	79.	**A**
20.	**C**	50.	**B**	80.	**D**
21.	**A**	51.	**C**	81.	**A**
22.	**D**	52.	**A**	82.	**C**
23.	**B**	53.	**B**	83.	**C**
24.	**C**	54.	**C**	84.	**B**
25.	**D**	55.	**C**	85.	**C**
26.	**C**	56.	**C**	86.	**B**
27.	**A**	57.	**D**	87.	**B**
28.	**C**	58.	**C**	88.	**C**
29.	**C**	59.	**A**	89.	**B**
30.	**D**	60.	**C**	90.	**B**

Physics

PART – I

SECTION – A

1. $\frac{v_1}{v_2} = \frac{27}{8}$

So, $\frac{m_1}{m_2} = \frac{8}{27}$ (from conservation of momentum)

$r \propto A^{1/3}$

$\frac{r_1}{r_2} = \left(\frac{8}{27}\right)^{1/3} = \frac{2}{3}$

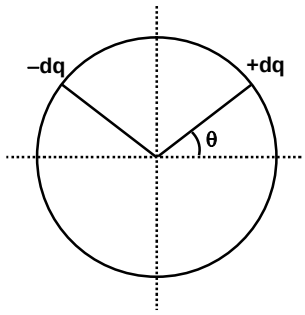
2. $7 \times \frac{D\lambda}{d} = \frac{(\mu_1 - \mu_2)t \cdot D}{d}$

$t = 9.1\ \mu m$

3. $dp = dq\ (2R \cos\theta)$

$= (3R \cos\theta\ d\theta)\ (2R \cos\theta)$

$p = \int dp = 48\ \pi \times 10^{-4}$ Coulomb-meter



4. In absorption spectrum electrons jumps from ground state to higher excited state.

5. $E \propto \frac{1}{n^2}$

$r \propto n^2$

$p \propto \frac{1}{n}$

$L \propto n$

6. $u = +15,\ \frac{1}{f} = \left(\frac{4}{3} - 1\right)\left(\frac{1}{\infty} - \frac{1}{10}\right)$

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} - \frac{1}{+15} = \frac{1}{-30}$

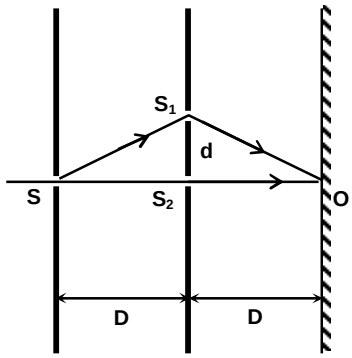
$v = 30$ cm

7. path difference = $(SS_1 + S_1O) - SO$

$= 2\sqrt{D^2 + d^2} - 2D = \dfrac{d^2}{D}$

$\dfrac{d^2}{D} = n\lambda$

$d = \sqrt{D\lambda}$ (n = 1)



8. Current through battery = $\dfrac{40}{5}$ = 8 amp

Current in branch CD = $\dfrac{1}{2}$ amp

So, $\dfrac{i_{CD}}{i_{AF}} = \dfrac{1/2}{8} = \dfrac{1}{16}$

9. $B_1 = \dfrac{\mu_0 N i r^2}{2(r^2 + h^2)^{3/2}}$

$B_2 = \dfrac{\mu_0 N i}{2r}$

10. Let plank is slight displacement a distance x.

$F_r = \dfrac{32}{3}kx$

$a = \dfrac{32k}{3m}x$

$\omega = \sqrt{\dfrac{32k}{3m}}$

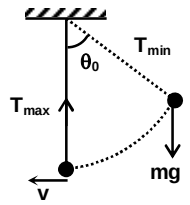
11. $T_{min} = mg\cos\theta_0$ …(i)

$T_{max} - mg = \dfrac{mv^2}{R}$ …(ii)

$\dfrac{1}{2}mv^2 = mg(R - R\cos\theta)$ …(iii)

$\dfrac{T_{max}}{T_{min}} = 4$

$\theta_0 = \cos^{-1}\left(\dfrac{1}{2}\right)$



12. Process PQ is isochoric
Process QR is isothermal
For process RP;

$V \propto \dfrac{1}{T}$

VT = constant

13. $2\left(L\frac{di_1}{dt}\right) = i_2R$

$2L\frac{i_0}{\tau_1}e^{-t/\tau_1} = R\cdot\frac{q_0}{\tau_2}e^{-t/\tau_2}$

$2e^{-t/\tau_1} = e^{-\frac{t}{\tau_2}}$

$t = \frac{\ell n2}{\frac{2R}{L}-\frac{1}{RC}}$

14. $q = 6e^{-2t}$

$i_C = 12\ e^{-2t}$

$i_L = i_1 + i_2 + i_C$

$i_L = 4 - 2e^{-2t}$

15. $i = \frac{V}{Z}$

$4 = \frac{200\sqrt{2}}{\sqrt{(50)^2+R^2}}$

$R = 50\ \Omega$

$V_R = i \times R = 200$ V

16. $20 = 10\log_{10}\left(\frac{I}{10^{-12}}\right)$

$I = 10^{-10}$ W/m^2 …(i)

$v = \sqrt{\frac{B}{\rho}} = 200$ m/s …(ii)

$(\Delta P_{max})^2 = 2I\rho v$

$\Delta P_{max} = 2\times10^{-4}$ N/m^2 …(iii)

$K = \frac{\omega}{v} = \frac{5}{2}m^{-1}$ ….(iv)

$\Delta P_{max} = BAK$

$A = 2 \times 10^{-9}$ m ….(v)

17. $\frac{\alpha_{eq}}{R_{eq}} = \frac{\alpha_1}{R_1} + \frac{\alpha_2}{R_2}$

$\frac{\alpha_{eq}}{800/3} = \frac{\alpha}{400} + \frac{4\alpha}{800}$

$\alpha_{eq} = 2\alpha$

18. $\oint\vec{E}\cdot ds = \frac{q_{in}}{\varepsilon_0} = 0$

Hence, $q_{in} = 0$

So, charge inside the surface must be zero.

19. from 0 to 4 sec, $e_1 = -\frac{3}{2}$ Volt

from 4 to 6 sec, $e_2 = +4$ Volt

$$e_{av} = \frac{\int e dt}{\int dt}$$

$$|e_{av}| = \frac{|-\frac{3}{2}\times 4 + 4\times 2|}{6} = \frac{1}{3} \text{ Volt}$$

20. $v = LT^{-1}$

$G = M^{-1}L^3T^{-2}$

$h = ML^2T^{-1}$

So, $\frac{v}{h} = \frac{1}{ML}$

and $\frac{G}{h^2} = \frac{1}{M^3L}$

Hence, $L = G^{1/2}, h^{1/2}, v^{-3/2}$

21. Displacement of point where force is applied

$$= \frac{4}{\sin 30^\circ} - \frac{4}{\sin 45^\circ}$$

$$= 4\left(2-\sqrt{2}\right) \text{meter}$$

So work done = $80\left(2-\sqrt{2}\right)$ Joule

22. $r^2 = b^2 + a^2e^2$ ...(1)

$b^2 = a^2\left(1-e^2\right)$ ...(2)

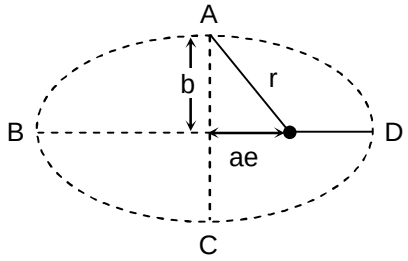
From equation (1) & (2)

$r = a$

Now from conservation of energy

$$-\frac{GMm}{2a} = \frac{1}{2}mv^2 - \frac{GMm}{a}$$

$$v = \sqrt{\frac{GM}{a}}$$

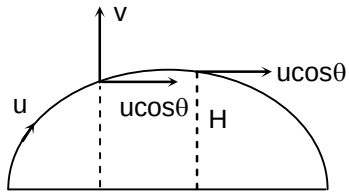


23. $u\cos\theta = \sqrt{\frac{3}{5}}\sqrt{v^2 + (u\cos\theta)^2}$

$$\frac{2}{5}u^2\cos^2\theta = \frac{3}{5}v^2 \quad \text{...(1)}$$

$$v^2 = u^2\sin^2\theta - 2g\frac{1}{3}\left(\frac{u^2\sin^2\theta}{2g}\right)$$

$$v^2 = \frac{2}{3}u^2\sin^2\theta \quad \text{...(2)}$$



24. $2(10\cos 60) = (2+2)v_0$

$v_0 = \frac{5}{2} m/s$ ...(1)

$4 \times \frac{5}{2} = (4+16)v$

$v = \frac{1}{2} m/s$ ...(2)

Now from conservation of energy

$\frac{1}{2} \times 4\left(\frac{5}{2}\right)^2 - \frac{1}{2}(4+16)\left(\frac{1}{2}\right)^2 = 4 \times 10 \times 1(1-\cos\theta)$

$\cos\theta = \frac{3}{4}$

$\theta = \cos^{-1}\left(\frac{3}{4}\right)$

25. $F_0 \cos \omega t - kA \cos \omega t = -mA\omega^2 \cos \omega t$

$\Rightarrow A = \frac{F_0}{k - m\omega^2}$

26. Heat produced $\propto r^2$ but change in temperature $\propto r$

27. Gravitation potential due to disc is

$V = -\frac{2Gm}{R^2}(\sqrt{R^2 + \ell^2} - \ell)$

So, $U_i = -\frac{Gmm}{\ell}$ $(R << \ell)$

$U_f = -\frac{2Gmm}{R}$ $(\ell \to 0)$

$\Delta U = U_i - U_f = Gmm\left(\frac{2}{R} - \frac{1}{\ell}\right)$

$\frac{1}{2}mv^2 + \frac{1}{2}mv^2 = Gmm\left(\frac{2}{R} - \frac{1}{\ell}\right)$

$v = \sqrt{Gm\left(\frac{2}{R} - \frac{1}{\ell}\right)}$

So, relative velocity

$v_R = 2v = \sqrt{4Gm\left(\frac{2}{R} - \frac{1}{\ell}\right)}$

28. $\log P = m \log V + C$ ...(i)

Where C is constant and m is slope

$m = \left[\frac{5.38 - 5.10}{3.10 - 3.30}\right] = -1.4$

So, from (i) $PV^{1.4} = K$ (constant)

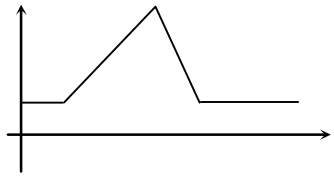
29. $\omega_H - \omega_L = \dfrac{R}{L}$

$\omega_H \times \omega_L = \omega_0^2$

$\Rightarrow \quad R = 750\,\Omega$

$\omega_0 = \dfrac{1}{\sqrt{LC}} \Rightarrow L = \dfrac{1}{\omega^2 C} = \dfrac{10^6}{10^4 \times 6} = 16.67H$

30. Output will be

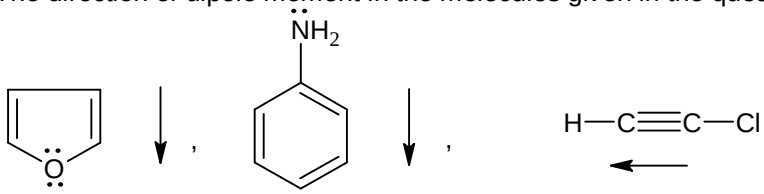


Chemistry

PART – II

SECTION – A

31. The direction of dipole moment in the molecules given in the question is



32. Pd belongs to 5th period and 10th group.

33. Highest energy electrons having n = 5 can lie in 5g sub-shell.
∴ Electronic configuration of element can be
$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 4f^{14}\ 5s^2\ 5p^6\ 5d^{10}\ 5f^{14}\ 5g^{18}\ 6s^2\ 6p^6\ 6d^{10}\ 7s^2\ 7p^6\ 8s^2$
∴ Maximum number of electrons can be 138.

34. $\pi = iCST$

$$\frac{74.4}{760} = 2 \times C \times 0.0821 \times 298$$

$$C = \frac{74.4}{760 \times 2 \times 0.0821 \times 298}$$

$$C = \frac{74.4}{37188.016} = 2 \times 10^{-3}\ \text{mol L}^{-1}$$

$K_{sp}\ (MX) = s^2 = (2 \times 10^{-3})^2 = 4 \times 10^{-6}$

35. (A) If x = 3, y = 0, then complex is $[Fe(Ox)_3]^{-3}$. This complex will show optical isomerism, not geometrical isomerism.
(B) If x = 1, y = 4, then complex is $[Fe(Ox)\ (H_2O)_4]^{+1}$. This complex neither shows geometrical nor optical.
(C) If x = 2, y = 2, then the complex is $[Fe(Ox)_2\ (H_2O)_2]^{-1}$. This complex can show geometrical and optical isomerism both.
(D) If x = 3 and y = 0, then complex will be $[Fe(Ox)_3]^{-3}$.

36. Three nucleotides are required to code each amino acid present in the chain. Therefore, chain having 21 amino acids will require 63 nucleotides to code the chain.

38. (i) PF_3 has greater bond angle due to back bonding.
(ii) LiCl has greater solubility in water due to its higher hydration enthalpy.
(iii) F – F is weaker bond due to lone pair – lone pair repulsion.
(iv) $\Delta_{eg}H$ order is : S > Se > O.

39. Xe has highest boiling point among the options given.

40. Correct statements are: a, b, d, f, g

41. Second order rate law for the reaction is
$r = K[A]^1[B]^1$
Since B is in excess, therefore new rate law for pseudo first order reaction becomes

$r = K_1[A]^1 \qquad \left[\text{Here } K_1 = K \times [B]^1\right]$

Now applying integrated equation for pseudo first order reaction

$$K_1 = \frac{2.303}{t}\log_{10}\frac{[A_0]}{[A]}$$

$$5 \times 10^{-3} \times 6 = \frac{2.303}{100}\log_{10}\frac{0.1}{[A]}$$

$$\frac{3 \times 10^{-2} \times 100}{2.303} = \log_{10}\frac{0.1}{[A]}$$

$$1.302 = \log_{10}\frac{0.1}{[A]}$$

Taking antilogarithm

$$20 = \frac{0.1}{[A]}$$

$$[A] = \frac{0.1}{20} = 5 \times 10^{-3}\ \text{M}$$

43. B is correct option.
d_{z^2} will be the highest energy orbital followed by $d_{xy} = d_{yz}$.

44. $B_2H_6 + 6H_2O \longrightarrow 2H_3BO_3 + 3H_2$

$1\text{g } B_2H_6 = \frac{1}{28}\text{mole} = 0.0357 \text{ mole } B_2H_6$

$\therefore$ Moles of H_3BO_3 produced = 0.0714 moles.
H_3BO_3 is a weak monobasic acid.

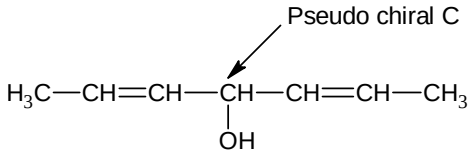
$$[H^+] = \sqrt{K_{a_1} \times C} = \sqrt{\frac{7.3 \times 10^{-10} \times 0.0714}{0.1}}$$

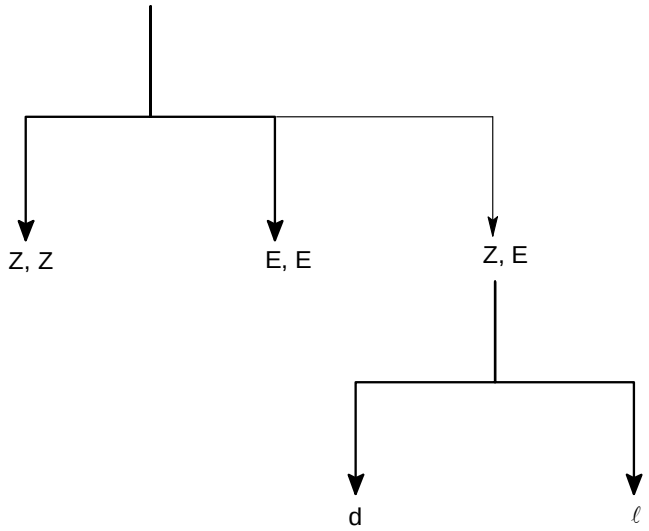
$$= \sqrt{0.521 \times 10^{-9}} = \sqrt{5.21 \times 10^{-10}}$$

$= 2.28 \times 10^{-5}$
Now pH = $-\log_{10} 228 \times 10^{-7}$
$= 7 - \log_{10} 228$
$= 7 - 2.357$
$= 4.64$

45. MnO_4^-, Ce^{+4}, $Cr_2O_7^{-2}$, CrO_4^{-2}, HNO_3, F_2 are very good oxidizing agents.

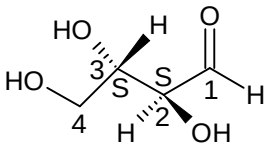
46.



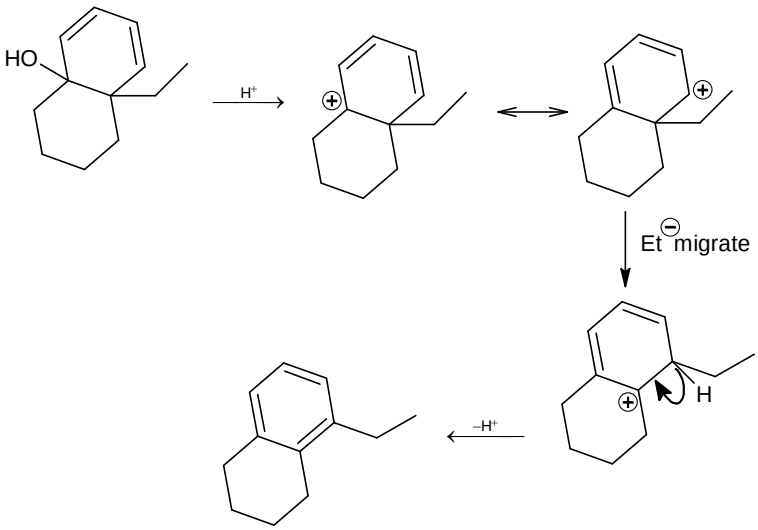


Thus, four stereoisomers are possible in total.

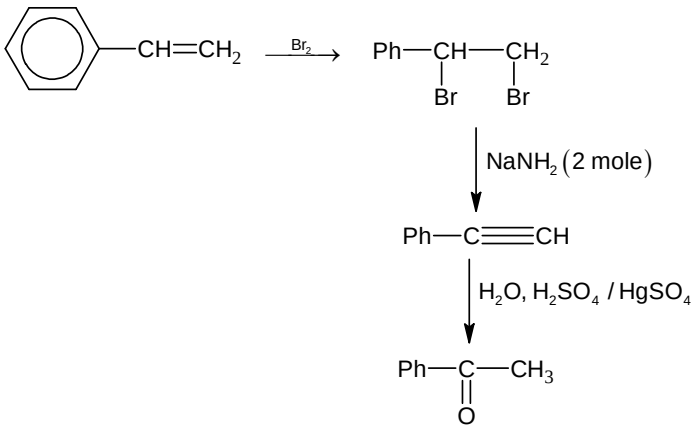
47. (I) is least reactive as C – Br bond acquires partial double bond character due to resonance. (II) is most reactive as it is allylic halide.

48. 

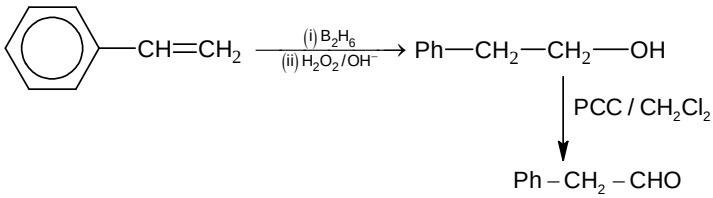
Both the chiral carbon have S, S-configuration. Therefore, correct option is B in which both the chiral carbon also have S, S-configuration.

49. 

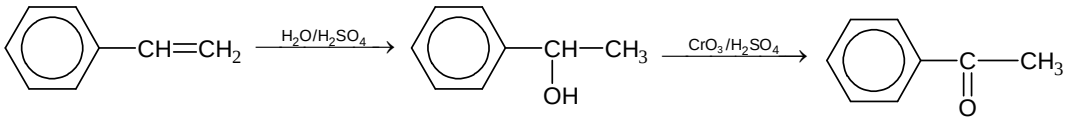
50. (A)



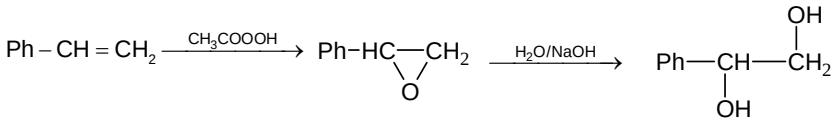
(B)



(C)



(D)



51. HCOOH, HCHO, $CH_3 - CH_2 - \underset{\underset{O}{\|}}{C} - CH_2OH$, H_3PO_2 give silver mirror test.

$CH_3 - C \equiv CH$ reacts with Ammonical silver nitrate but does not give metallic silver.

52. (i) Decolourize bromine in CCl_4] $\Rightarrow$ Evidence of presence of multiple bond.
(ii) Absorbed hydrogen catalytically] $\Rightarrow$ Evidence of presence of multiple bond.
(iii) Gives precipitate with ammonical cuprous chloride] $\Rightarrow$ Evidence of terminal alkyne.
(iv) 448 mL = 1. 49 g

22400 mL = $\frac{1.49}{448} \times 22400 = 74.5$ g (molar mass)

$\therefore$ Option (A) is correct.

53. (II), (V), (VI), (VIII), (IX) and (X) are antiseptic.

54. Nylon-6 is a condensation homopolymer.
Nylon-6,6 is a condensation heteropolymer.
Bakelite is a condensation heteropolymer.
PMMA is an addition polymer.

55. Diborane shows unsymmetrical Cleavage with NH_3.

56. Ce^{+3} is more stable than Ce^{+4}.

57. $Ni(NO_3)_2$ solution is electrolysed by using Ni electrode. Therefore, no change in molarity of solution after electrolysis.

58. Order of acidic strength of conjugate acid of species given in question

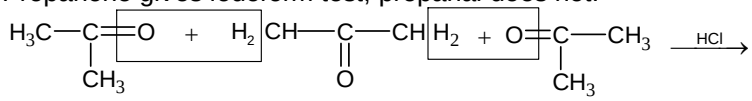
$HI > H_2S > NH_4^+ > R-\overset{+}{N}H_3$

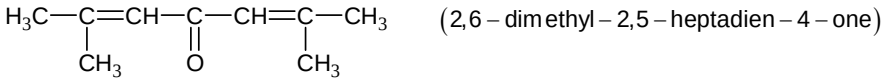
$\therefore$ Proton accepting tendency of conjugate bases will be

$R-NH_2 > NH_3 > \bar{S}H > I^-$

59. - I effect of – OMe group at m-position decreases stability of diazonium ion and hence increases its reactivity.

60. Propanone gives iodoform test, propanal does not.





Mathematics

PART – III

SECTION – A

61. $y^n - 1 = (y - 1)(y - \lambda_1)(y - \lambda_2) ..... (y - \lambda_{n-1})$

$\Rightarrow \frac{y^n - 1}{y - 1} = (y - \lambda_1)(y - \lambda_2).....(y - \lambda_{n-1})$

$\Rightarrow 1 + y + y^2 + ... + y^{n-1} = (y - \lambda_1)(y - \lambda_2) ..... (y - \lambda_{n-1})$

Putting $y = 1$

$\Rightarrow (1 - \lambda_1)(1 - \lambda_2) ..... (1 - \lambda_{n-1}) = n$

62. $\left|\frac{z_3 - z_1}{z_3 - z_2}\right| = |i|$

$|z_3 - z_1| = |z_3 - z_2|$

Also $\arg\left(\frac{z_3 - z_1}{z_3 - z_2}\right) = \frac{\pi}{2}$

$\therefore$ Angle at vertex z_3 is 90°

63. If exactly one negative then the value of expression is 1
If exactly two negative then value of expression is –1
If all three negative then value of expression is –3
If all three positive then value of expression is 3

64. $|P|\,|Q|\,|R| = 3$

$|Q| = \frac{3}{4}$

65. $\cos^2\alpha + p + \sin^2\alpha + p = -4$

$p = -\frac{5}{2}$

$(\cos^2\alpha + p)(\sin^2\alpha + p) = \frac{61}{16}$

$\sin^2\alpha\cos^2\alpha = \frac{1}{16}$

$x^2 - (\tan^2\alpha + \cot^2\alpha)x + \tan^2\alpha\cot^2\alpha = 0$

Required equation $x^2 - 14x + 1 = 0$

66. $\alpha + \beta = -p$, $\alpha\beta = +q$

$\because$ α, β are the roots of the equation $x^{2n} + p^n x^n + q^n = 0$

$\Rightarrow \alpha^n, \beta^n$ are the roots of the quadratic equation $y^2 + p^n y + q^n = 0$

$\alpha^n + \beta^n = -p^n$, $\alpha^n\beta^n = q^n$

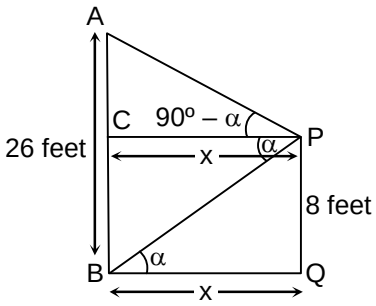
$$f\left(\frac{\alpha}{\beta}\right) = \frac{\left(1 + \frac{\alpha}{\beta}\right)^n}{1 + \frac{\alpha^n}{\beta^n}} = \frac{(\alpha + \beta)^n}{\alpha^n + \beta^n} = \frac{(-p)^n}{-p^n} = -1$$

67. $\tan\alpha = \frac{8}{x}$

$\tan(90^o - \alpha) = \cot\alpha = \frac{18}{x}$

$\frac{x}{8} = \frac{18}{x}$

$x = 12$



68. $\sin 2\alpha = -\frac{1}{2}$

i.e., $\alpha = 105^o, 165^o, 285^o, 345^o$

69. Let a and b be roots of $x^2 - 7x + 8 = 0$ then $a + b = 7$ and $ab = 8$ and $\angle C = 60^o$

$a^2 + b^2 = (a + b)^2 - 2ab = 49 - 16 = 33$

$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$

$\frac{1}{2} = \frac{33 - c^2}{16}$

$c^2 = 33 - 8 = 25 \Rightarrow c = 5$

$r.R = \frac{abc}{2(a+b+c)} = \frac{8 \times 5}{2 \times 12} = \frac{5}{3}$

70. AM $\geq$ G.M.

$\frac{\alpha^8 + \beta^8 + 1 + 1 + 1 + 1 + 1 + 1}{8} \geq \left(\alpha^8\beta^8\right)^{1/8}$

$\alpha^8 + \beta^8 + 6 \geq 8\alpha\beta$

Which is possible only if $\alpha^8 = \beta^8 = 1$

$\Rightarrow \alpha = \pm 1$ and $\beta = \pm 1$

71. $\sum_{i=1}^{4000} \frac{1}{p_i p_{i+1}} = \frac{1}{d}\left[\left(\frac{1}{p_1} - \frac{1}{p_2}\right) + \left(\frac{1}{p_2} - \frac{1}{p_3}\right) + ... + \left(\frac{1}{p_{4000}} - \frac{1}{p_{4001}}\right)\right] = 10$

$\frac{p_{4001} - p_1}{d(p_1 \cdot p_{4001})} = 10 \Rightarrow \frac{4000\,d}{d(p_1 \cdot p_{4001})} = 10 \Rightarrow p_1\, p_{4001} = 400$ and $p_1 + p_{4001} = 50$

$|p_1 - p_{4001}| = 30$

72. $R(b + c) = a\sqrt{bc} = 2R \sin A \sqrt{bc}$

$\sin A = \frac{b + c}{2\sqrt{bc}}$

Now $\sin A \leq 1 \Rightarrow \frac{b + c}{2\sqrt{bc}} \leq 1$

$\left(\sqrt{b} - \sqrt{c}\right)^2 \leq 0 \Rightarrow b = c$

$\sin A = 1 \Rightarrow \angle A = \frac{\pi}{2}$ and $b = c$

Hence, triangle is right angled isosceles triangle

73. $2\lambda_1 + 2\lambda_2 + 3\lambda_3 - 1 = 0$
$2\lambda_1 + 2\lambda_2 + 3\lambda_3 = 1$
$\therefore \left|2\lambda_1 + 2\lambda_2 + 3\lambda_3\right| \le \sqrt{4+4+9}\sqrt{\lambda_1^2 + \lambda_2^2 + \lambda_3^2}$
$\lambda_1^2 + \lambda_2^2 + \lambda_3^2 \ge \frac{1}{17}$

74. Any point on the first line is $P(3r_1 + 1, r_1 + 2, 2r_1 + 3)$ and on second line is $Q(r_2 + 3, 2r_2 + 1, 3r_2 + 2)$ P and Q represent the same point if $r_1 = r_2 = 1$ and the point of intersection of the given lines is P(4, 3, 5), the planes given is (A), (B), (C) and (D) all passes through P the plane at greatest distance is one which is at a distance equal to OP from the origin is $\sqrt{4^2 + 3^2 + 5^2} = \sqrt{50}$ which is satisfies by the plane (B)

75. By the change of origin there is no change in the variance but by the change of scale. New variance $|\alpha|^2 \sigma^2$ here $|\alpha| = 2$

76. ${}^7C_4(2)^4(-1)^3$

77. Equation of director circle of $x^2 + y^2 = 25$ is $x^2 + y^2 = 50$
$k \in \left(-5\sqrt{2}, -5\right) \cup \left(5, 5\sqrt{2}\right)$

78. Given below

q	~q	p	~p	~q ∧ p	p ∨ ~p	(~q ∧ p) ∨ (p ∨ ~p)
T	F	T	F	F	T	T
T	F	F	T	F	T	T
F	T	T	F	T	T	T
F	T	F	T	F	T	T

79. Let the number of elements in set P, Q, R, S be p, q, r, s respectively now to find positive integral solutions of $p + q + r + s = 12$, $p, q, r, s > 0$ and $p < q$
$\Rightarrow$ Total ways = 70

80. Equation of tangent to $y^2 = 4x$ at (4, 4) is $x - 2y + 4 = 0$
Let the other point of intersection is $(t^2, 2t)$
$\Rightarrow$ Equation of ellipse is $(2x - y - 4)(2x - (t + 2)y + 4t)) + \lambda(x - 2y + 4)(2x - (t - 1)y - 2t) = 0$
Since axes parallel to coordinate axes
$\Rightarrow$ Coefficient of $xy = 0 \Rightarrow t = -3$

81. $\left(a^2 - 3a - 3\right)^2 \ge \left(a^2 + 7a - 13\right)^2$
$\Rightarrow 20(a^2 + 2a - 8)(a - 1) < 0$
$\Rightarrow (a + 4)(a - 2)(a - 1) < 0$
$\Rightarrow a \in (-\infty, -4) \cup (1, 2)$
Probability = $\frac{1}{11}$

82. As $2q = p + r \Rightarrow$ p and r both are even or both are odd
If p and r both odd then number of ways = $2 \cdot {}^5C_2$
If p and r both even then number of ways = $2 \cdot {}^4C_2 + {}^4C_1$
$\therefore$ Total ways = 36

83. On adding all equation a + b + c = 6.4
Now subtracting one by one all three equation, we get a = 2, b = 1.3, c = 3.1

84. $xf'(x) + 2f(x) = 3x$
I.F. = $e^{2\int \frac{1}{x}dx} = x^2$
$x^2f(x) = c + x^3$
c = 0
f(x) = x
f(3) = 3

85. Let $I_1 = \int_0^1 t^{1004}(1-t)^{1004}\,dt$
$I_2 = \int_0^1 t^{1004}(1-t^{2010})^{1004}\,dt$
Put $t^{1005} = u \Rightarrow 1005t^{1004}\,dt = du$
$= \frac{1}{1005}\int_0^1 (1-u^2)^{1004}\,du$
Using property $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$
$= \frac{1}{1005}\int_0^1 u^{1004}(2-u)^{1004}\,du$
Put u = 2v

86. Let $g(x) = ax^2 + bx + c$ then g(3) = 9, and $\lim_{x\to 2}\frac{g(x)}{x-2} = 7$

87. $\lim_{t\to\lambda} f(t) = \frac{3}{2}$ and $\lim_{t\to\lambda} g(t) = \frac{1}{2}$; $\lim_{t\to\lambda} f(t)\cdot g(t) = \frac{3}{4}$

88. $\lim_{n\to\infty}\sum_{r=0}^{n}\frac{{}^nC_r}{n^r}\int_0^1 x^{r+2}dx = \int_0^1 x^2\left(\lim_{n\to\infty}\sum_{r=0}^{n}{}^nC_r\left(\frac{x}{n}\right)^r\right)dx$
$= \int_0^1 x^2\left(\lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n\right)dx = \int_0^1 x^2\,e^x dx = e-2$

89. $\frac{1}{1-\sin(\cos^{-1}x)} = 2$
$\Rightarrow \sin(\cos^{-1}x) = \frac{1}{2}$
$\Rightarrow x = \frac{\sqrt{3}}{2}$

90. Let P(5 cos θ, 4 sin θ)
PN = 4 sin θ ⇒ AN = 5 – 5 cos θ
⇒ A′N = 5 + 5 cos θ
$\Rightarrow \frac{PN^2}{AN\cdot A'N} = \frac{16\sin^2\theta}{25(1-\cos^2\theta)} = \frac{16}{25}$